Technical Report: Analysis of National Travel Trends and a Car-Following Simulation

To: Federal Highway Administration (FHWA)

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Introduction

As traffic increases in urban areas, the Federal Highway Administration (FHWA) requires data analysis to improve traffic safety and efficiency. The goal of this project was to analyze the National Household Travel Survey (NHTS) to understand the current composition of the U.S. vehicle fleet, and second, to utilize the Next Generation Simulation (NGSIM) dataset to validate the Intelligent Driver Model (IDM) for predicting vehicle following behavior.

Methods

Variable Selection

I selected the following variables for analysis:

- vehicle_type & vehicle_age: Selected to identify how long different classifications of vehicles remain operating
- fuel_type: Selected to observe the transition from gas run cars to alternative fuel sources.
- leader_position & follower_speed: Taken from NGSIM to provide the inputs (gap and velocity) required for the car-following simulation.

Plot Construction

- Bar Graph: Constructed by grouping data by fuel_type and calculating the mean vehicle_age to compare vehicle longevity across different power sources.
- Histogram: Generated using 25 bins to show the frequency distribution of vehicle ages across the survey sample.
- Time-Series Line Plots: Created to track speed and gap distance over time for a specific vehicle trajectory (Trajectory #2 was chosen for my analysis).
- IDM Acceleration Plot: A comparative plot putting actual NGSIM acceleration with simulated acceleration generated by the IDM.

Computational Methods

The team utilized Euler's Method for integration to simulate the follower vehicle. By calculating the IDM acceleration at each 0.1-1 second interval (dt), I updated the velocity (v) and position (x) using the following equations:

$$v_{next} = v_{current} + a_{IDM} * dt$$

$$x_{next} = x_{current} + v_{next} * dt$$

Results and Discussion

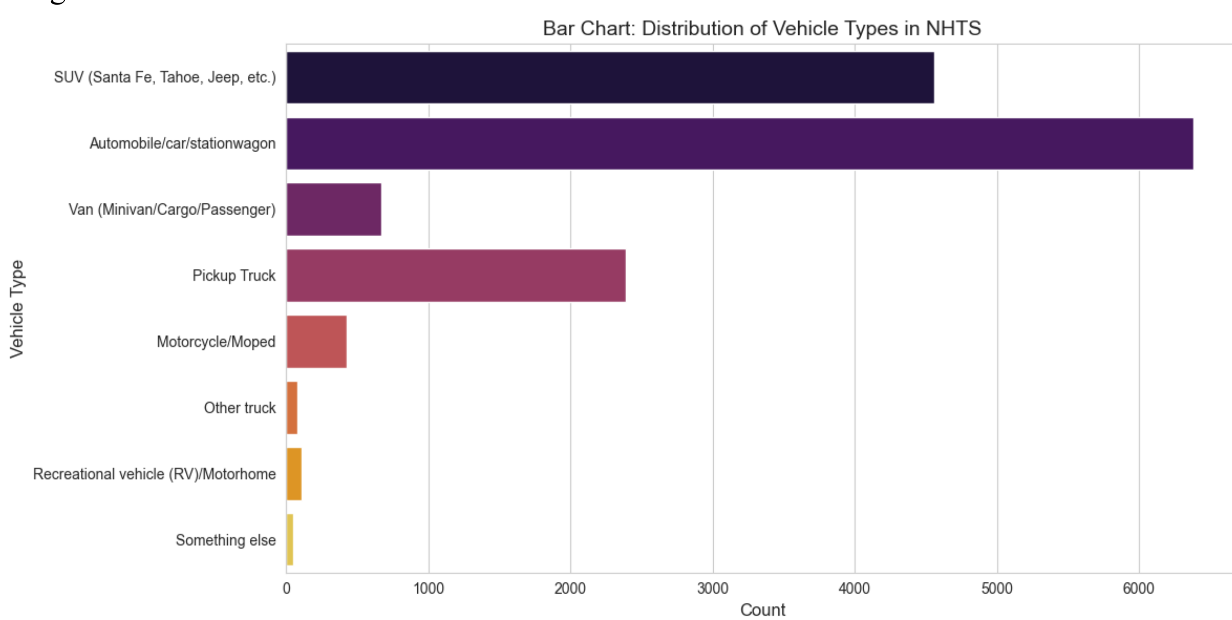
Fleet Distribution and Longevity

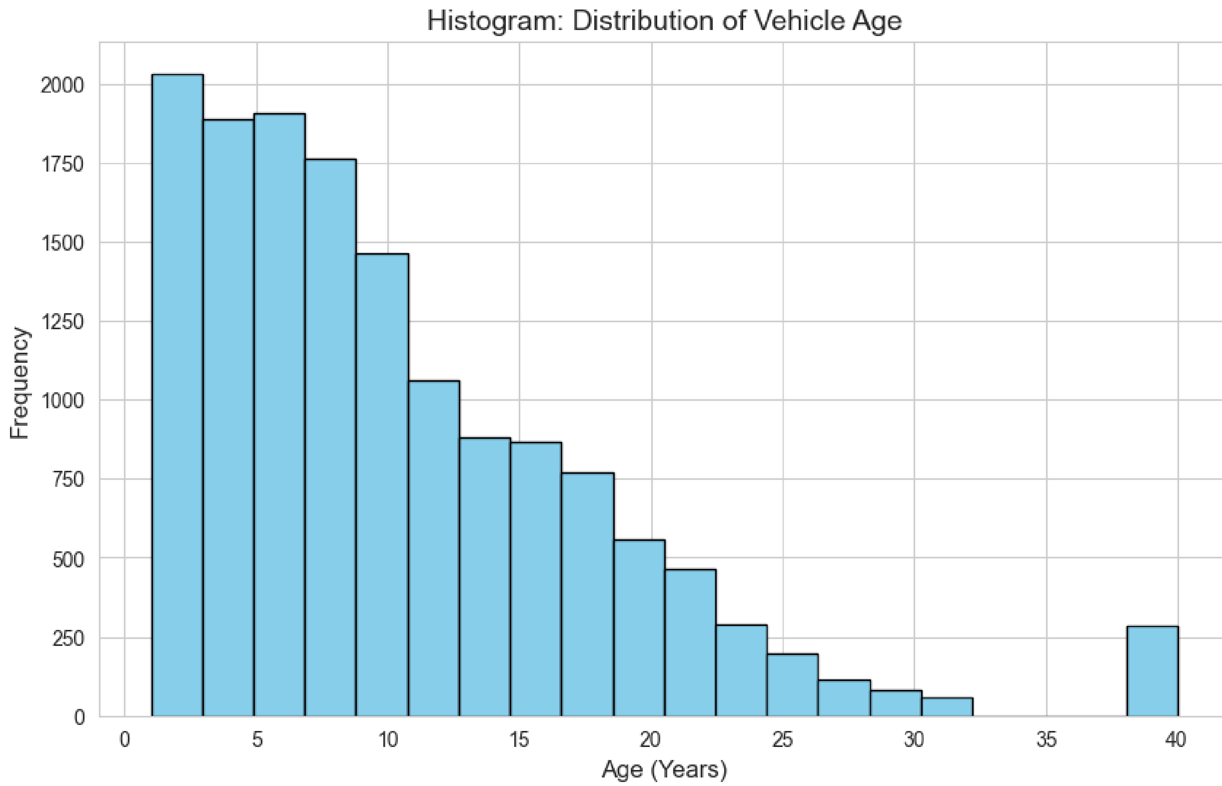
The Histogram of Vehicle Age shows a right-skewed distribution, with a large peak in vehicles aged 2–10 years and a large drop-off after 15 years, confirming that the majority of the U.S. vehicles are relatively modern. The Boxplot specifically highlights that Electric and Hybrid vehicles have much lower median ages (under 5 years) and tighter spreads compared to Diesel and other vehicles, which show significantly higher age variance and older outliers. This is due to the fact that electric vehicles are fairly new and in turn cannot have much variance or outliers because there is only a small range of time it could possibly span from.

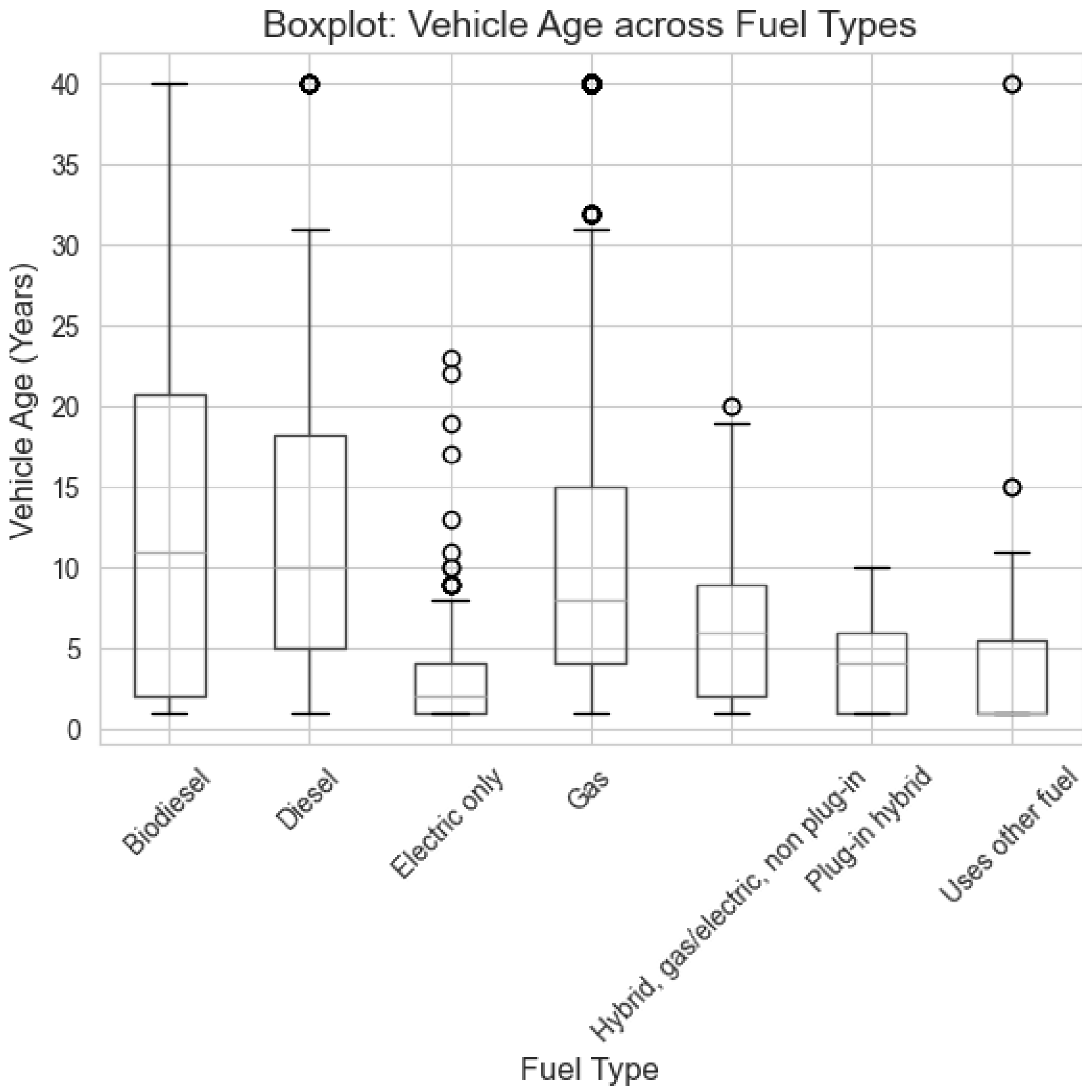
IDM Simulation Accuracy

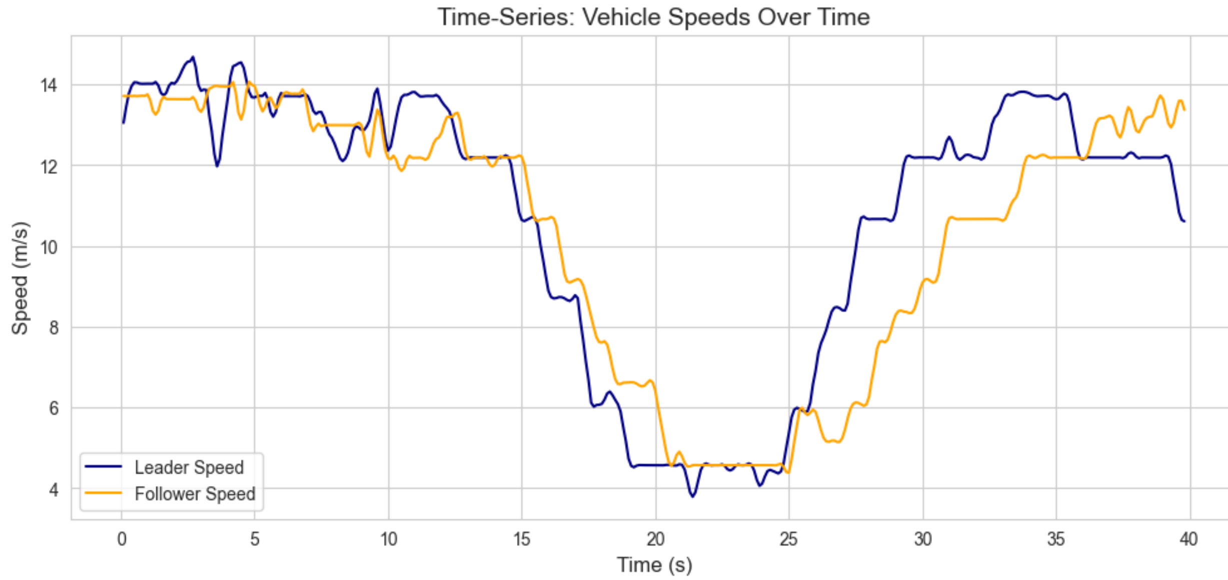
The Acceleration Comparison plot shows that while the Simulated IDM (red line) successfully captures the general flow of traffic, it acts as a smoother. Real-world human drivers (green line) exhibit inconsistent acceleration/braking between -4 and +4 m/s², however the IDM remains fairly constant, rarely exceeding +1 or -2 m/s². The Speed Time-Series confirms this, showing the follower reacting to the leader's deceleration starting at the 15-second mark with a slight time lag.

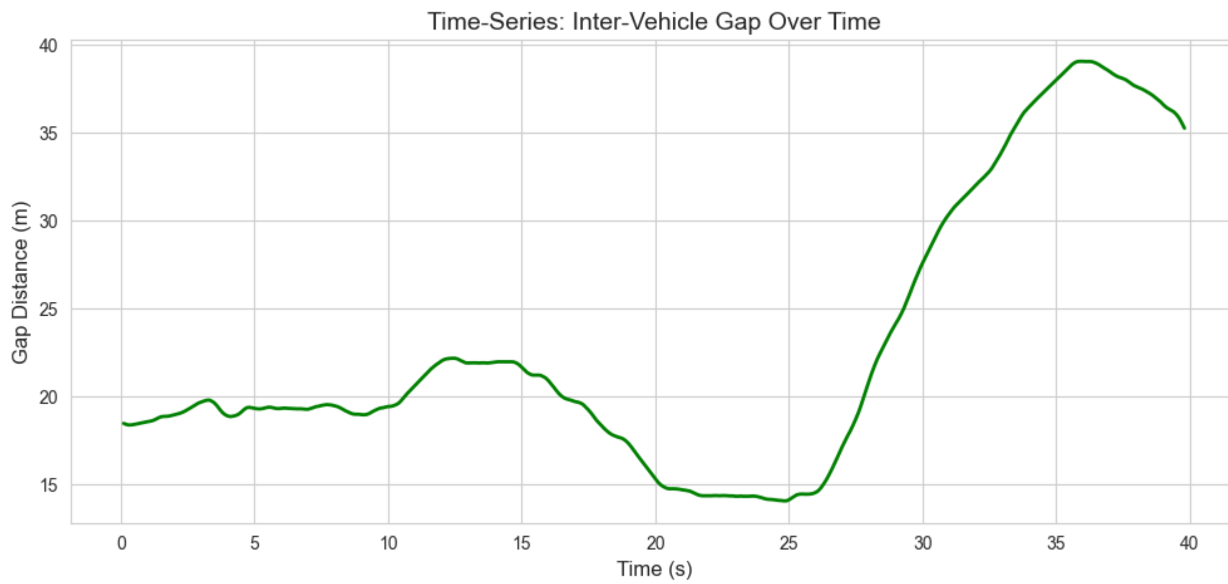
Images of Results from Code:

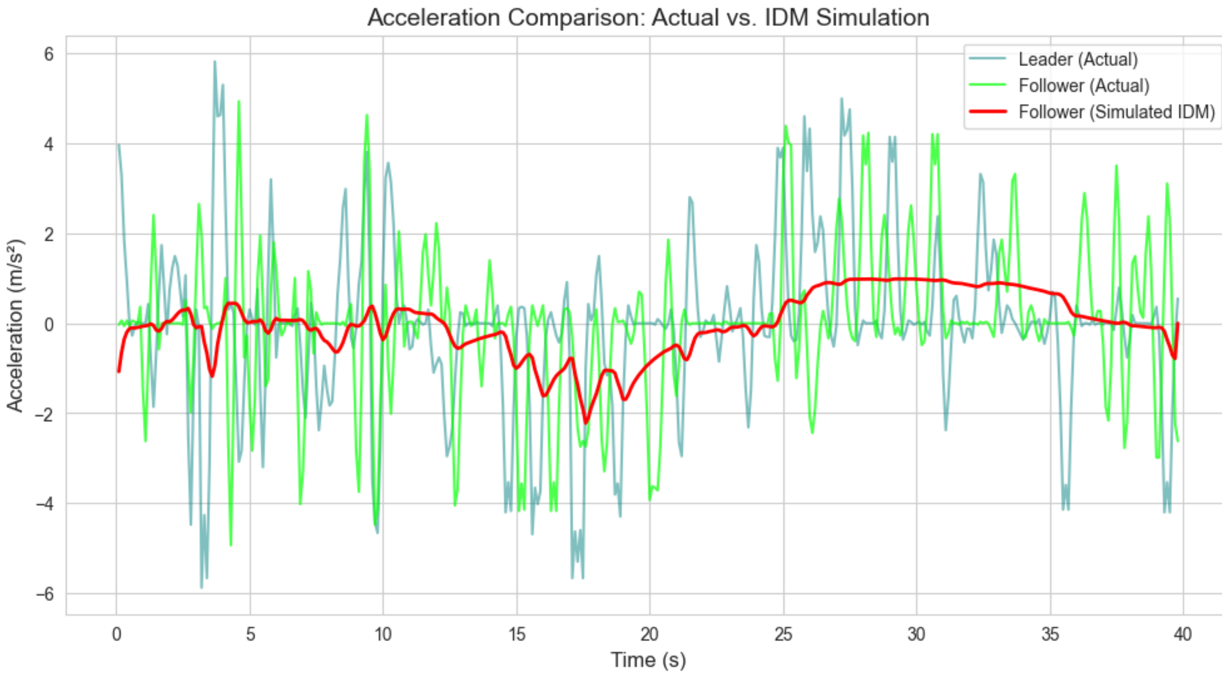



Histogram: Distribution of Vehicle Age
Frequency
2000
1750
1500
1250
1000
750
500
250
0
0
5
10
15
20
25
30
35
40
Age (Years)


Boxplot: Vehicle Age across Fuel Types
Vehicle Age (Years)
0
5
10
15
20
25
30
35
40
Biodiesel
Diesel
Electric only
Gas
Hybrid, gas/electric, non plug-in
Plug-in hybrid
Uses other fuel
Fuel Type


Time-Series: Vehicle Speeds Over Time
Speed (m/s)
Time (s)
Leader Speed
Follower Speed
4
6
8
10
12
14
0
5
10
15
20
25
30
35
40


Time-Series: Inter-Vehicle Gap Over Time
Gap Distance (m)
Time (s)
15
20
25
30
35
40
0
5
10
15
20
25
30
35
40



References

Federal Highway Administration (FHWA). (2022). *National Household Travel Survey (NHTS) Data User Guide*.

NGSIM. (2006). *Next Generation Simulation: Interstate 80 Vehicle Trajectories Dataset*.